

Data Mining – Lecture 2

1- A/an _____ represents an entity .

- a- Data object
- b- Attribute
- c- Both
- d- None

Answer : a. Note (“synonyms : sample , example , instance and others depends on the field “)

2- An attribute is a data field which represents a characteristic or feature of a data object.

- a- True
- b- False

Answer : a. Note “ another synonyms , dimension , feature and variable , depends on the field”

3- Attribute vector is a set of attributes that describe an object.

- a- True
- b- False

Answer : a.

4- Nominal attributes has only two values “states”.

- a- True
- b- False

Answer : a. note (“its qualitative”)

5- Nominal _____ , states are not equally important .

- a- Symmetric
- b- Asymmetric
- c- Both
- d- None

Answer : b.

6- At Nominal symmetric both states are equally valuable and have the same weight .

- a- True
- b- False

Answer : a.

7- At _____ attributes , values have a meaningful order or ranking but magnitude between successive values is not known .

- a- Ordinal
- b- Binary
- c- Numeric
- d- All above

Answer : a. examples (“ professional rank , grade ...)

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8- ____ is an example of Qualitative attribute types.

- a- Ordinal
- b- Binary
- c- Nominal
- d- All above

Answer : d.

9-At ____ numeric attributes are measured on a scale of equal size units .

- a- Interval scaled
- b- Ratio scaled
- c- Both
- d- None

Answer : a. Notes (“ it doesn’t have a true zero point and not possible to be expressed as multiples “)

10- At the ratio scaled which has a true zero point ,, the value can be expressed as a multiple of another .

- a- True
- b- False

Answer : a.

11- Salary , weight and years of experiences are examples of ____ .

- a- Interval scaled
- b- Ratio scaled
- c- Both
- d- None

Answer : b , note (“ to understand more , review the previous couple of questions. “)

12- ____ attribute has a finite or countably infinite set of values, integers or otherwise.

- a- Discrete
- b- Continuous
- c- Both
- d- None

Answer : a.

13- ____ means a chop of extreme values at both ends .

- a- Trim
- b- Combine
- c- Cut
- d- None

Answer : a.

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14- ____ is the value which occurs most frequently in attribute values .

- a- Mean
- b- Median
- c- Mode
- d- None

Answer : c.

15- At ____ data observations tend to be very close to the mean .

- a- High Standard deviation
- b- Low Standard deviation
- c- Zero
- d- None

Answer : b.

16- Data is spread out over a large range of values at high standard deviation.

- a- True
- b- False

Answer : a.

17- Range is the difference between the largest and the smallest values .

- a- True
- b- False

Answer : a .

18- Interquartile Range is difference between Q3 and Q2.

- a- True
- b- False

Answer : b , explanation (“Between Q3 and Q1 which gives us the middle area “)

19- ____ represents , Median (Q2) , quartiles Q1 , Q3 and smallest and largest individual observations in order .

- a- Five number summary
- b- Boxplots
- c- Both
- d- None

Answer : a . explanation (“ boxplots must be related to the visualization of that technique “)

20- Boxplots is a visualization technique for the five-number summary .

- a- True
- b- False

Answer : a.

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21- Whiskers equals to $1.5 \times \text{IQR}$.

- a- True
- b- False

Answer : a .

Age	23	23	27	27	39	41	47	49	50	52	54	54	56	57	58	58	60	61
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

- Draw the boxplots for age and %fat.

The following questions related to the following example (Age) not fat

22- Find the median .

- a- 51
- b- 50
- c- 23
- d- 52

Answer : a .

23- Find Q1 and Q3.

- a- 39,50
- b- 39,57
- c- 57,39
- d- None

Answer : b . explanation (Q1 is the median of first Half and Q3 is the median of the second half)

24-IQR = ____.

- a- 51
- b- 18
- c- 27
- d- None

Answer : b. explanation (" IQR =Q3-Q1")

25- For the Age , the boxplot is ____.

- a- Symmetric data
- b- Negatively skewed
- c- Positively skewed
- d- None

Answer : b.

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- 26- ____ : each pair of values is treated as a pair of coordinates and plotted as points in plane .
- a- Histogram
 - b- Scatter Plots
 - c- Boxplot
 - d- None

Answer : b.

- 27- Statistical descriptions are about ____, while Similarity and dissimilarity is about ____.
- a- Attributes , Variables
 - b- Attributes , Objects
 - c- Objects , attributes
 - d- None

Answer : b. note (similarity and dissimilarity measures objects proximity)

- 28- Dissimilarity matrix stores a collection of proximities for all pairs of n objects as an n by n matrix

- a- True
- b- False

Answer : a.

PROXIMITY MEASURES BINARY ATTRIBUTES - EXAMPLE

name	gender	fever	cough	Test-1	Test-2	Test-3	Test-4
Ali	M	Y	N	P	N	N	N
Adam	M	Y	Y	N	N	N	N
Selma	F	Y	N	P	N	P	N
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Y and P → 1

N and N → 0

The following questions related to this question. Note(gender is the only symmetric)

“We will assume we only use asymmetric attributes .

Remember that will use this rule : *Asymmetric* $\rightarrow d(i,j) = \frac{r+s}{q+r+s}$

q is # attributes that equal 1 for i & j

r is # attributes that equal 1 for i but 0 for j

s is # attributes that equal 0 for i but 1 for j

t is # attributes that equal 0 for i & j

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29- Lets find the $d(\text{Ali}, \text{Adam}) = \underline{\hspace{1cm}}$.

- a- 0.67
- b- 0.33
- c- 0.75
- d- 0

Answer : a.

Explanation (Based on the formula we need to find (r, s and q) basically if you found r and s you've found 4 of 5 . so lets begin with r.

- a- r is equal to 1 because r is from the attributes that equal to 1 for the 'i' which is ali and 0 for 'j' which is Adam . Look at cases which help us only which is "Test 1 "
- b- s is equal to 1 as well be cause at "cough" Ali is negative while Adam is positive so we just have one case here so it's 1 .
- c- now for q which is only 1 when both of attributes are 1 which at our case is once at "fever" which both are Y.

30- Find the $d(\text{Ali}, \text{Selma}) = \underline{\hspace{1cm}}$.

- a- 0.67
- b- 0.33
- c- 0.75
- d- 0

Answer : b .

Explanation : (alrright now we have more understanding it will be brief)

- a- r is equal to 0 because Ali has no cases where he is 1 and Selma is 0
- b- s is equal to 1 because we got one case where Selma is 1 while Ali is 0
- c- q is equal to 2 here because ! Ali and Selma where 1 at two cases " Fever and Test 1 "

"Thanks :D , please if there is any questions or mistakes here just dm please "